

PEGDA hydrogels: load-bearing strands from rheology, compared with FLICK junction fluctuations

PEGDA 5K / 10K / 20K · G' digitised from the Origin G' - Q plots · $\rho = 1.12 \text{ g cm}^{-3}$, $T = 298 \text{ K}$

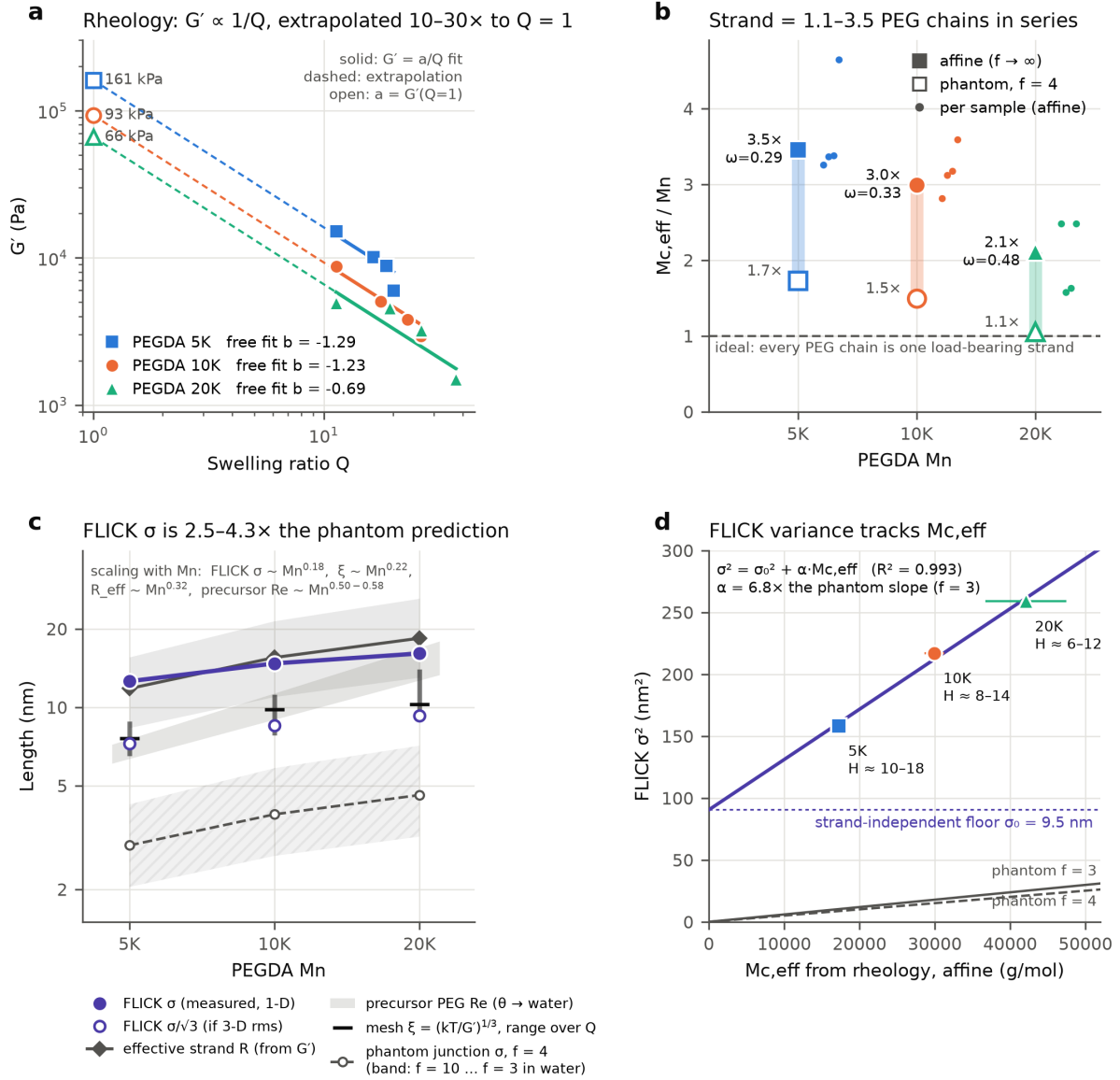


Figure 1 | Rheology and FLICK read the same load-bearing strand, but through different averages. (a) Storage modulus G' against swelling ratio Q for PEGDA 5K (blue squares), 10K (orange circles) and 20K (green triangles), log-log. Solid lines: least-squares fits of $G' = a/Q$ (exponent fixed at -1) over the measured range. Dashed lines: extrapolation of those fits to $Q = 1$. Open symbols: the extrapolated dry-network modulus $a = G'(Q = 1)$, labelled in kPa. The legend gives the exponent b of an unconstrained fit $G' = aQ^b$. **(b)** Effective strand mass relative to the precursor, $M_{c,eff}/M_n = \rho RT/(a \cdot M_n)$. Filled symbols: affine model ($f \rightarrow \infty$). Open symbols: phantom model with $f = 4$. The shaded bar spans the two. Small dots: the same quantity for each individual sample, $\rho RT/(G' \cdot Q \cdot M_n)$. $\omega = M_n/M_{c,eff}$ is the fraction of PEG chains that carry load (affine). The dashed line marks the ideal network in which every PEG chain is one load-bearing strand. **(c)** All length scales against M_n , log-log. Violet filled circles: FLICK σ , read as a per-axis junction displacement. Violet open circles: $\sigma/\sqrt{3}$, the value if FLICK σ is a 3-D rms. Diamonds: unperturbed size R_0 of the effective strand from $M_{c,eff}$ (θ statistics, affine). The band around them runs from phantom- θ to affine-in-water. Grey band: precursor PEG end-to-end distance R_e (θ to water). Black bars: rubber-elastic mesh size $\xi = (kT/G')^{1/3}$, showing the range across each Q series (tick = median). Open circles with dashed line: per-axis junction fluctuation that a phantom network ($f = 4$, θ) predicts from the same G' . The hatched band runs from $f = 10$ (θ) to $f = 3$ (in water), which is the largest the model allows. **(d)** FLICK variance σ^2 against the rheological $M_{c,eff}$ (affine). Horizontal bars show the fit standard error. Violet line: $\sigma^2 = \sigma_0^2 + \alpha \cdot M_{c,eff}$. Dotted line: the strand-independent floor σ_0^2 . Grey lines: phantom predictions through the origin (solid $f = 3$, dashed $f = 4$). $H = (\sigma_{FLICK}/\sigma_{phantom, f=4})^2$ is given as a range from water to θ strand statistics.

What the figure shows, and how the panels build the argument

Focus. PEGDA networks have no controlled junction functionality f and contain loops, dangling chains and dense poly(acrylate) clusters. So the question is not the 'true' strand length. It is whether a macroscopic measurement (G') and a microscopic one (FLICK junction fluctuation) describe the same *effective, load-bearing* strand, and what their disagreement means. The figure argues that **both report the same effective strand and agree on how it changes with M_n , but differ 3-4× in magnitude**. That difference is expected when one method averages stiffness and the other averages compliance, so it measures network heterogeneity.

Logic arc through the panels

a • Establish the input. Within each molecular weight G' falls roughly as $1/Q$. Extrapolating to $Q = 1$ gives a dry-network modulus a for each gel. The panel is honest about the leverage: the extrapolation reaches 10–30× beyond the data. It also shows that the 20K series is the weakest ($R^2 = 0.63$).

b • Turn the modulus into network meaning. a converts to an effective strand mass. The load-bearing strand is 1–3.5 PEG chains long, so only about 30–50 % (affine) to 60–95 % (phantom) of PEG chains carry load. The network is least defective at 20K. The affine-phantom bar is the unavoidable model bracket when f is unknown. Every later number inherits it.

c • Put every length on one axis and expose the mismatch. Once the precursor size, effective strand size, mesh size, predicted junction fluctuation and FLICK σ share one axis, the key fact is visible. The phantom model, fed the same G' , predicts junction fluctuations of 3–7 nm. FLICK measures 12.6–16 nm, which is 2.5–4.3× larger even at the model's upper limit. The M_n scaling is informative too. FLICK ($M_n^{0.18}$) follows the mesh size ($M_n^{0.22}$), not the precursor chain ($M_n^{0.5-0.58}$).

d • Resolve the mismatch. Plotted against the rheological $M_{c,eff}$, FLICK σ^2 falls on a straight line ($R^2 = 0.993$; 0.92 against M_n). So FLICK responds to the same effective strand that G' counts: the two agree on the trend. The line has two parts. A strand-independent floor $\sigma_0 \approx 9.5$ nm, and a slope about 7× the phantom value: the two disagree on magnitude. A modulus is a stiffness average (k) dominated by the percolating backbone. A fluctuation is a compliance average ($1/k$) dominated by loosely tethered junctions. Since $\langle k \rangle \langle 1/k \rangle \geq 1$ for any heterogeneous network, $H = (\sigma_{FLICK}/\sigma_{phantom})^2$ works as a heterogeneity index (5K 10–18, 10K 8–14, 20K 6–12). The M_n trend in H depends on what σ_0 is. Measuring the FLICK localisation floor on an immobilised reference settles it.

Take-home. Rheology: the average load-bearing strand is 1–3.5 PEG chains. FLICK: junctions move as if held by strands roughly 3–9× longer, so most tracked junctions sit in softer-than-average surroundings. The gap between the two methods is a measure of network heterogeneity, not a contradiction.

Supporting notes: equations, results, caveats and next steps

analysis.py holds every input, equation and number. results.md has the full tables, and pegda_rheology_vs_flick.png / .pdf is the figure. Re-run with python3 analysis.py after changing the inputs at the top of the script.

What rheology can and cannot tell you

A modulus counts **elastically effective (load-bearing) strands**, v_e , times kT , times a model prefactor. It does not see loops, dangling PEG chains, sol, or how the junctions are arranged. Every "strand length" derived from G' is therefore an **effective** number. This is the effective, conflated quantity you described.

Quantity	Equation	What it needs
Effective strand density (swollen state)	$v_e = G'/kT$	nothing (model-free up to the affine or phantom prefactor)
Rubber-elastic mesh size	$\xi = (kT/G')^{1/3}$	nothing
Dry-state strand density	$G' = v_{e,dry} RT \phi \lambda^2$, with $\phi = 1/Q$ and $\lambda^2 = (\phi_0/\phi)^{2/3}$	the preparation fraction ϕ_0
Effective strand mass	$M_{c,eff} = (1-2/f) \cdot \rho RT / (G' Q \lambda^2)$	f, ϕ_0
Strand size	$R_0 = (\langle R^2 \rangle / M \cdot M_c)^{1/2}$, PEO θ : 0.805 Å ² mol/g; water: $R_g = 0.0215 M^{0.583}$ nm	chain statistics
Phantom junction fluctuation	$\langle \delta r^2 \rangle = (f-1)/[f(f-2)] \cdot R_0^2$; per axis $\sigma^2 = \langle \delta r^2 \rangle / 3$	f, R_0

Your $G' = a/Q$ fit ($b = -1$) is the $\lambda = 1$ case: each gel is taken as relaxed in the state where it was measured. I kept that convention so the numbers compare directly with your tables. If G' was measured on equilibrium-swollen gels made at $\phi_0 > \phi$, every $M_{c,eff}$ below rises by $(\phi_0 Q)^{2/3}$. For example, 20 wt% and $Q = 15$ gives a factor of about 1.9.

Two corrections to the equations on your slide

- 1 **Drop Flory's $(1 - 2M_c/M_n)$ factor.** It corrects for free ends when long primary chains are cross-linked at random ($M_n \gg M_c$). In PEGDA the PEG chain is the strand ($M_c \approx M_n$), so the factor goes to zero or negative.
- 2 **An affine network has no junction fluctuations.** The affine model pins junctions to the macroscopic deformation, so $\sigma_{affine} = 0$. The " $\sigma(\text{Affine})$ " column in your table is $R_e/\sqrt{6}$, which is the strand's radius of gyration, not a fluctuation. Only the phantom (or constrained-junction) model predicts a fluctuation.

Results ($\rho = 1.12 \text{ g cm}^{-3}$, $T = 298 \text{ K}$)

	5K	10K	20K
$a = G'(Q=1)$ from the a/Q fit (Pa)	$160,600 \pm 10,000$	$92,800 \pm 4,100$	$66,000 \pm 8,400$
R^2 of the fit; free exponent b	0.88; -1.29	0.96; -1.23	0.63; -0.69
$M_{c,eff}$ affine ($f \rightarrow \infty$)	17.3k	29.9k	42.1k
$M_{c,eff}$ phantom $f = 4$	8.6k	15.0k	21.0k
$M_{c,eff} / M_n$ (affine to phantom)	3.5 to 1.7	3.0 to 1.5	2.1 to 1.05
Load-bearing fraction $\omega = M_n/M_c$ (affine)	0.29	0.33	0.48
Precursor R_e (θ / water), nm	6.3 / 7.6	9.0 / 11.3	12.7 / 16.9
Effective strand R_0 (θ , affine), nm	11.8	15.5	18.4
Mesh $\xi = (kT/G')^{1/3}$, nm (range over the Q series)	6.5-8.8	7.8-11.2	9.4-14.0
Phantom junction σ per axis, $f = 4$ (θ / water), nm	2.9 / 3.9	3.9 / 5.4	4.6 / 6.5
Largest possible phantom σ ($f = 3$, water), nm	4.2	5.8	7.1
FLICK σ, nm	12.6	14.7	16.1

The phantom σ uses the M_c that the same G' implies for that f , so it depends on G' and f only. Because of this, $\sigma^2 = C \cdot M_{c,aff} \cdot (f-1)/(3f^2)$ cannot exceed its $f = 3$ value for any $f \geq 3$.

The story

1. The rheology is internally consistent but "sees" defective networks.

$G' \propto Q^{-1}$ to $Q^{-1.3}$ holds within each molecular weight. Your fixed- $b = -1/3$ fits fail for a reason: each series changes the preparation concentration as well as the swelling, so it does not test how a single network swells. The load-bearing strand works out to **1-3.5 PEG chains in series**, equivalently **only 30-50 % (affine) to 60-95 % (phantom) of PEG chains carry load**. 5K is the most defective, which fits the known picture: loops and dense poly(acrylate) clusters are most common at high acrylate density. $M_{c,eff} \propto M_n^{0.64}$, not M_n^1 . The effective strand grows more slowly than the precursor chain.

2. Length-scale ladder (panel c). From smallest to largest:

- phantom junction fluctuation: 3-7 nm
- mesh ξ : 7-14 nm
- precursor R_e and effective-strand R : 6-26 nm
- FLICK σ : 12.6-16 nm

FLICK is 2.5-4.3× larger than any phantom prediction the same G' allows. Even if σ is a 3-D rms rather than a per-axis value, it is still about 1.5-2.3× larger.

3. FLICK still tracks the rheological strand (panel d).

FLICK σ^2 is linear in $M_{c,eff}$ ($R^2 = 0.993$, against 0.92 when plotted against M_n). The fit splits into two parts:

- $\sigma^2 = \sigma_0^2 + \alpha \cdot M_{c,eff}$
- **$\sigma_0 \approx 9.5$ nm**: a floor that does not depend on the strand. Candidates are localization precision, drift, or motion of a whole cluster or heterogeneity region.
- The slope α is **6.8× the phantom slope** per axis, or 2.3× if σ is 3-D.

So FLICK and rheology agree on the *trend*. FLICK responds to the same effective strand that G' counts, not to the precursor length. They disagree on the *magnitude*.

4. Why the magnitude should disagree: stiffness average vs compliance average.

- G' is a **stiffness average**, $\langle k \rangle$, and is dominated by the percolating, well-connected backbone.
- A junction's thermal fluctuation is a **compliance average**, $\langle 1/k \rangle$, and is dominated by weakly tethered junctions in soft, defect-rich regions.
- For any heterogeneous network, $\langle k \rangle \langle 1/k \rangle \geq 1$ (Jensen's inequality).

The ratio **$H = (\sigma_{\text{FLICK}} / \sigma_{\text{phantom}})^2 \approx \langle k \rangle \langle 1/k \rangle$** therefore works as a **heterogeneity index**:

	5K	10K	20K
H (water to θ strands)	10-18	8-14	6-12

H falls as M_n rises. 5K PEGDA looks the most heterogeneous, which matches its lowest load-bearing fraction ω . Those two readouts are independent: one comes from G' and one from FLICK.

Caveat: this trend in H comes from the σ_0 floor. If you subtract σ_0 first, H becomes nearly the same for all three molecular weights (about 8 for $f = 4$, set by α). Pinning down what σ_0 is (see check 2 below) therefore decides whether the heterogeneity trend is real.

One-line take-home

- Rheology says the average load-bearing strand is 1–3.5 PEG chains.
- FLICK says junctions move as if they were held by strands roughly 3–9× longer (phantom model, $f = 3-4$), meaning most tracked junctions sit in softer-than-average surroundings.
- The gap between the two measures network heterogeneity. It appears to shrink from 5K to 20K, but that trend depends on what σ_0 is.

Things to check in the input data

- **The 20K " $G'(Q=1)$ " in your table is 64,091, but your own Origin fit says 65,091.** M_c for 20K is 42.0k, not 42.7k.
- **All three molecular weights have their first point at $Q = 11.31$,** identical to 0.01. That looks like a copy or paste error. Please verify it.
- **The 20K series is non-monotonic in $G' \cdot Q$** ($R^2 = 0.63$). Its $M_{c,eff}$ carries a ± 13 % error, and its free exponent is -0.69 .
- **A consistency test (the c^* -theorem):** if G' was measured at swelling equilibrium, then $G' \approx \Pi_{mix}(Q)$ for every molecular weight, so all three series should fall on one $G'(Q)$ curve. At $Q = 11.3$ they differ by 3×. This suggests G' was measured in the as-prepared state, or that Q and G' refer to different states. That decides whether $\lambda = 1$ (the convention used here) is right.
- **Precursor R_e :** your 4.52 / 6.40 / 9.05 nm set corresponds to $\langle R^2 \rangle / M = 0.41 \text{ \AA}^2 \text{ mol/g}$, about half the standard PEO θ value of 0.805 (Rubinstein & Colby). Your 7.55 / 11.31 / 16.94 nm set is $\sqrt{6} \cdot R_g$ for PEG in water.
- **Keep one σ convention.** Your precursor σ is $R_e/\sqrt{3}$, which is a per-axis spread. Your " $\sqrt{6}\sigma$ " column instead treats σ as R_g . The script treats FLICK σ as the per-axis rms displacement of a junction, and shows the 3-D-rms alternative alongside.

What would make the numbers reliable rather than effective

- 1 **Record ϕ_0 (precursor wt%) for each sample and the state in which G' was measured.** This fixes λ and removes the largest systematic error, up to about $2\times$ on M_c .
- 2 **Measure the FLICK localization floor** on an immobilized reference, such as a dried or glassy gel or dye on glass. If it is about 9 nm, σ_0 is instrumental. If it is smaller, σ_0 is a real heterogeneity length.
- 3 **Measure the sol fraction and acrylate conversion** (by extraction and by NMR or FTIR). This separates "dangling or unreacted" chains from "loops", which gives a bound on ω that does not depend on the model.
- 4 **Look at the FLICK *distribution*** of σ across junctions, not just its mean. For a phantom network it should be narrow. A broad, long-tailed distribution directly confirms the stiffness-versus-compliance explanation.